Formal_Analysis

2026-03-26

```r
set.seed(1234)
data<-read.csv("~/Desktop/STAT4052/Final Project/D2D2016FoodStudy_data.csv")
str(data)
```

```
## 'data.frame':    2660 obs. of  55 variables:
##  $ person_id      : int  10001 10001 10001 10001 10001 10001 10001 10001 10002 10002 ...
##  $ percent_discard: int  1 3 90 0 92 91 91 96 49 0 ...
##  $ beef           : int  1 1 1 1 0 0 0 0 1 1 ...
##  $ class1         : int  0 0 0 0 0 0 0 0 1 1 ...
##  $ level2         : int  0 0 0 0 0 0 0 0 1 0 ...
##  $ level3         : int  0 0 1 0 1 0 0 0 0 0 ...
##  $ bestby         : int  0 0 0 1 1 0 0 1 0 0 ...
##  $ useby          : int  0 0 1 0 0 0 0 0 1 1 ...
##  $ date_days      : int  1 3 1 2 7 1 7 3 1 3 ...
##  $ price          : num  4.79 12.79 4.79 6.39 2.49 ...
##  $ large          : int  0 1 0 0 0 0 1 0 1 0 ...
##  $ age            : num  49.5 49.5 49.5 49.5 49.5 49.5 49.5 49.5 69.5 69.5 ...
##  $ gender         : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ educ           : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ income         : num  6.25 6.25 6.25 6.25 6.25 6.25 6.25 6.25 4.25 4.25 ...
##  $ hhld_size      : int  2 2 2 2 2 2 2 2 1 1 ...
##  $ child_pres     : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ nonwhite       : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ preshop_1      : int  4 4 4 4 4 4 4 4 1 1 ...
##  $ preshop_2      : int  4 4 4 4 4 4 4 4 1 1 ...
##  $ preshop_3      : int  4 4 4 4 4 4 4 4 1 1 ...
##  $ preshop_4      : int  4 4 4 4 4 4 4 4 2 2 ...
##  $ instore_5      : int  4 4 4 4 4 4 4 4 1 1 ...
##  $ instore_6      : int  4 4 4 4 4 4 4 4 3 3 ...
##  $ instore_7      : int  4 4 4 4 4 4 4 4 3 3 ...
##  $ tosspre_8      : int  5 5 5 5 5 5 5 5 2 2 ...
##  $ tosspre_9      : int  5 5 5 5 5 5 5 5 1 1 ...
##  $ tosspre_10     : int  5 5 5 5 5 5 5 5 5 5 ...
##  $ tosspre_11     : int  5 5 5 5 5 5 5 5 3 3 ...
##  $ tosspre_12     : int  5 5 5 5 5 5 5 5 5 5 ...
##  $ tosspre_13     : int  5 5 5 5 5 5 5 5 5 5 ...
##  $ tosspre_14     : int  2 2 2 2 2 2 2 2 4 4 ...
##  $ tosspost_15    : int  3 3 3 3 3 3 3 3 4 4 ...
##  $ tosspost_16    : int  3 3 3 3 3 3 3 3 2 2 ...
##  $ tosspost_17    : int  4 4 4 4 4 4 4 4 4 4 ...
##  $ tosspost_18    : int  2 2 2 2 2 2 2 2 5 5 ...
##  $ tosspost_19    : int  2 2 2 2 2 2 2 2 1 1 ...
##  $ amount_20      : int  2 2 2 2 2 2 2 2 2 2 ...
##  $ amount_21      : int  2 2 2 2 2 2 2 2 2 2 ...
```

```
##  $ amount_22     : int  2 2 2 2 2 2 2 2 3 3 ...
##  $ amount_23     : int  2 2 2 2 2 2 2 2 0 0 ...
##  $ amount_24     : int  0 0 0 0 0 0 0 0 1 1 ...
##  $ amount_25     : int  2 2 2 2 2 2 2 2 1 1 ...
##  $ amount_26     : int  1 1 1 1 1 1 1 1 2 2 ...
##  $ amount_27     : int  1 1 1 1 1 1 1 1 2 2 ...
##  $ skill_28      : int  5 5 5 5 5 5 5 5 4 4 ...
##  $ skill_29      : int  5 5 5 5 5 5 5 5 3 3 ...
##  $ skill_30      : int  5 5 5 5 5 5 5 5 5 5 ...
##  $ skill_31      : int  5 5 5 5 5 5 5 5 4 4 ...
##  $ skill_32      : int  5 5 5 5 5 5 5 5 5 5 ...
##  $ skill_33      : int  5 5 5 5 5 5 5 5 5 5 ...
##  $ family_34     : int  2 2 2 2 2 2 2 2 4 4 ...
##  $ family_35     : int  5 5 5 5 5 5 5 5 4 4 ...
##  $ compost_36    : int  5 5 5 5 5 5 5 5 5 5 ...
##  $ recycle_37    : int  5 5 5 5 5 5 5 5 1 1 ...
```

```r
head(data)
```

```
##   person_id percent_discard beef class1 level2 level3 bestby useby date_days
## 1     10001               1    1      0      0      0      0     0         1
## 2     10001               3    1      0      0      0      0     0         3
## 3     10001              90    1      0      0      1      0     1         1
## 4     10001               0    1      0      0      0      1     0         2
## 5     10001              92    0      0      0      1      1     0         7
## 6     10001              91    0      0      0      0      0     0         1
##   price large  age gender educ income hhld_size child_pres nonwhite preshop_1
## 1  4.79     0 49.5      0    0   6.25         2          0        0         4
## 2 12.79     1 49.5      0    0   6.25         2          0        0         4
## 3  4.79     0 49.5      0    0   6.25         2          0        0         4
## 4  6.39     0 49.5      0    0   6.25         2          0        0         4
## 5  2.49     0 49.5      0    0   6.25         2          0        0         4
## 6  1.99     0 49.5      0    0   6.25         2          0        0         4
##   preshop_2 preshop_3 preshop_4 instore_5 instore_6 instore_7 tosspre_8
## 1         4         4         4         4         4         4         5
## 2         4         4         4         4         4         4         5
## 3         4         4         4         4         4         4         5
## 4         4         4         4         4         4         4         5
## 5         4         4         4         4         4         4         5
## 6         4         4         4         4         4         4         5
##   tosspre_9 tosspre_10 tosspre_11 tosspre_12 tosspre_13 tosspre_14 tosspost_15
## 1         5          5          5          5          5          2           3
## 2         5          5          5          5          5          2           3
## 3         5          5          5          5          5          2           3
## 4         5          5          5          5          5          2           3
## 5         5          5          5          5          5          2           3
## 6         5          5          5          5          5          2           3
##   tosspost_16 tosspost_17 tosspost_18 tosspost_19 amount_20 amount_21 amount_22
## 1           3           4           2           2         2         2         2
## 2           3           4           2           2         2         2         2
## 3           3           4           2           2         2         2         2
## 4           3           4           2           2         2         2         2
## 5           3           4           2           2         2         2         2
## 6           3           4           2           2         2         2         2
```

```
##   amount_23 amount_24 amount_25 amount_26 amount_27 skill_28 skill_29 skill_30
## 1         2         0         2         1         1        5        5        5
## 2         2         0         2         1         1        5        5        5
## 3         2         0         2         1         1        5        5        5
## 4         2         0         2         1         1        5        5        5
## 5         2         0         2         1         1        5        5        5
## 6         2         0         2         1         1        5        5        5
##   skill_31 skill_32 skill_33 family_34 family_35 compost_36 recycle_37
## 1        5        5        5         2         5          5          5
## 2        5        5        5         2         5          5          5
## 3        5        5        5         2         5          5          5
## 4        5        5        5         2         5          5          5
## 5        5        5        5         2         5          5          5
## 6        5        5        5         2         5          5          5
```

```r
summary(data)
```

```
##    person_id     percent_discard       beef            class1      
##  Min.   :10001   Min.   :  0.00   Min.   :0.0000   Min.   :0.0000  
##  1st Qu.:10084   1st Qu.:  0.00   1st Qu.:0.0000   1st Qu.:0.0000  
##  Median :10167   Median :  9.00   Median :0.0000   Median :0.0000  
##  Mean   :10167   Mean   : 27.02   Mean   :0.4429   Mean   :0.4323  
##  3rd Qu.:10250   3rd Qu.: 50.00   3rd Qu.:1.0000   3rd Qu.:1.0000  
##  Max.   :10333   Max.   :100.00   Max.   :1.0000   Max.   :1.0000  
##                                                                    
##      level2           level3          bestby           useby       
##  Min.   :0.0000   Min.   :0.000   Min.   :0.0000   Min.   :0.0000  
##  1st Qu.:0.0000   1st Qu.:0.000   1st Qu.:0.0000   1st Qu.:0.0000  
##  Median :0.0000   Median :0.000   Median :0.0000   Median :0.0000  
##  Mean   :0.2462   Mean   :0.244   Mean   :0.3391   Mean   :0.3289  
##  3rd Qu.:0.0000   3rd Qu.:0.000   3rd Qu.:1.0000   3rd Qu.:1.0000  
##  Max.   :1.0000   Max.   :1.000   Max.   :1.0000   Max.   :1.0000  
##                                                                    
##    date_days         price            large             age       
##  Min.   :1.000   Min.   : 1.990   Min.   :0.0000   Min.   :21.00  
##  1st Qu.:1.000   1st Qu.: 2.990   1st Qu.:0.0000   1st Qu.:29.50  
##  Median :3.000   Median : 4.990   Median :1.0000   Median :49.50  
##  Mean   :2.926   Mean   : 6.341   Mean   :0.5071   Mean   :45.41  
##  3rd Qu.:3.000   3rd Qu.: 7.990   3rd Qu.:1.0000   3rd Qu.:59.50  
##  Max.   :7.000   Max.   :15.990   Max.   :1.0000   Max.   :84.50  
##                                                                   
##      gender            educ             income          hhld_size   
##  Min.   :0.0000   Min.   :0.0000   Min.   : 1.850   Min.   :1.00  
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.: 4.250   1st Qu.:2.00  
##  Median :0.0000   Median :1.0000   Median : 8.750   Median :2.00  
##  Mean   :0.3274   Mean   :0.6669   Mean   : 9.159   Mean   :2.64  
##  3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:12.500   3rd Qu.:4.00  
##  Max.   :1.0000   Max.   :1.0000   Max.   :25.000   Max.   :7.00  
##                                                                   
##    child_pres        nonwhite        preshop_1       preshop_2    
##  Min.   :0.0000   Min.   :0.0000   Min.   :1.000   Min.   :1.000  
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:2.000   1st Qu.:2.000  
##  Median :0.0000   Median :0.0000   Median :3.000   Median :4.000  
##  Mean   :0.2406   Mean   :0.1414   Mean   :2.956   Mean   :3.342  
```

```
##  3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:4.000   3rd Qu.:4.000
##  Max.   :1.0000   Max.   :1.0000   Max.   :5.000   Max.   :5.000
##
##    preshop_3        preshop_4        instore_5        instore_6
##  Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
##  1st Qu.:3.000   1st Qu.:4.000   1st Qu.:3.000   1st Qu.:2.000
##  Median :4.000   Median :4.000   Median :4.000   Median :3.000
##  Mean   :3.845   Mean   :4.028   Mean   :3.524   Mean   :3.215
##  3rd Qu.:5.000   3rd Qu.:5.000   3rd Qu.:4.000   3rd Qu.:4.000
##  Max.   :5.000   Max.   :5.000   Max.   :5.000   Max.   :5.000
##                                  NA's   :95      NA's   :16
##    instore_7        tosspre_8        tosspre_9        tosspre_10
##  Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
##  1st Qu.:3.000   1st Qu.:2.000   1st Qu.:2.000   1st Qu.:4.000
##  Median :4.000   Median :4.000   Median :3.000   Median :4.000
##  Mean   :3.582   Mean   :3.398   Mean   :3.114   Mean   :4.138
##  3rd Qu.:4.000   3rd Qu.:4.000   3rd Qu.:4.000   3rd Qu.:5.000
##  Max.   :5.000   Max.   :5.000   Max.   :5.000   Max.   :5.000
##  NA's   :47
##    tosspre_11       tosspre_12       tosspre_13       tosspre_14
##  Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
##  1st Qu.:2.000   1st Qu.:2.000   1st Qu.:2.000   1st Qu.:1.000
##  Median :2.000   Median :4.000   Median :2.000   Median :1.000
##  Mean   :2.747   Mean   :3.492   Mean   :2.816   Mean   :1.701
##  3rd Qu.:4.000   3rd Qu.:5.000   3rd Qu.:4.000   3rd Qu.:2.000
##  Max.   :5.000   Max.   :5.000   Max.   :5.000   Max.   :5.000
##
##   tosspost_15      tosspost_16    tosspost_17      tosspost_18      tosspost_19
##  Min.   :1.000   Min.   :1.00   Min.   :1.000   Min.   :1.000   Min.   :1.000
##  1st Qu.:2.000   1st Qu.:2.00   1st Qu.:2.000   1st Qu.:3.000   1st Qu.:1.000
##  Median :4.000   Median :3.00   Median :4.000   Median :4.000   Median :2.000
##  Mean   :3.337   Mean   :3.33   Mean   :3.547   Mean   :3.842   Mean   :2.065
##  3rd Qu.:5.000   3rd Qu.:4.00   3rd Qu.:5.000   3rd Qu.:5.000   3rd Qu.:2.000
##  Max.   :5.000   Max.   :5.00   Max.   :5.000   Max.   :5.000   Max.   :5.000
##
##    amount_20        amount_21        amount_22        amount_23
##  Min.   :0.000   Min.   :0.000   Min.   :0.000   Min.   :0.000
##  1st Qu.:2.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000
##  Median :2.000   Median :2.000   Median :1.000   Median :2.000
##  Mean   :2.457   Mean   :1.715   Mean   :1.676   Mean   :1.956
##  3rd Qu.:3.000   3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:3.000
##  Max.   :5.000   Max.   :5.000   Max.   :5.000   Max.   :5.000
##
##    amount_24        amount_25        amount_26        amount_27
##  Min.   :0.000   Min.   :0.000   Min.   :0.000   Min.   :0.000
##  1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000
##  Median :2.000   Median :1.000   Median :1.000   Median :1.000
##  Mean   :1.935   Mean   :1.496   Mean   :1.667   Mean   :1.316
##  3rd Qu.:3.000   3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:2.000
##  Max.   :5.000   Max.   :5.000   Max.   :5.000   Max.   :5.000
##
##     skill_28         skill_29         skill_30         skill_31
##  Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
##  1st Qu.:4.000   1st Qu.:3.000   1st Qu.:4.000   1st Qu.:4.000
```

```
##  Median :4.000   Median :4.000   Median :4.000   Median :4.000
##  Mean   :4.127   Mean   :3.794   Mean   :4.055   Mean   :4.053
##  3rd Qu.:5.000   3rd Qu.:5.000   3rd Qu.:5.000   3rd Qu.:5.000
##  Max.   :5.000   Max.   :5.000   Max.   :5.000   Max.   :5.000
##
##     skill_32        skill_33      family_34       family_35       compost_36
##  Min.   :2.000   Min.   :2.00   Min.   :1.000   Min.   :1.000   Min.   :1.000
##  1st Qu.:4.000   1st Qu.:4.00   1st Qu.:3.000   1st Qu.:3.000   1st Qu.:4.000
##  Median :4.000   Median :4.00   Median :4.000   Median :4.000   Median :5.000
##  Mean   :4.108   Mean   :4.27   Mean   :3.423   Mean   :3.707   Mean   :4.366
##  3rd Qu.:5.000   3rd Qu.:5.00   3rd Qu.:4.000   3rd Qu.:4.000   3rd Qu.:5.000
##  Max.   :5.000   Max.   :5.00   Max.   :5.000   Max.   :5.000   Max.   :5.000
##
##    recycle_37
##  Min.   :1.000
##  1st Qu.:1.000
##  Median :1.000
##  Mean   :2.003
##  3rd Qu.:2.000
##  Max.   :5.000
##
```

```
dim(data)
```

```
## [1] 2660   55
```

```
data$beef<-as.factor(data$beef)
data$class1<-as.factor(data$class1)
data$level2<-as.factor(data$level2)
data$level3<-as.factor(data$level3)
data$bestby<-as.factor(data$bestby)
data$useby<-as.factor(data$useby)
data$large<-as.factor(data$large)
data$gender<-as.factor(data$gender)
data$educ<-as.factor(data$educ)
data$child_pres<-as.factor(data$child_pres)
data$nonwhite<-as.factor(data$nonwhite)
str(data)
```

```
## 'data.frame':    2660 obs. of  55 variables:
##  $ person_id      : int  10001 10001 10001 10001 10001 10001 10001 10001 10002 10002 ...
##  $ percent_discard: int  1 3 90 0 92 91 91 96 49 0 ...
##  $ beef           : Factor w/ 2 levels "0","1": 2 2 2 2 1 1 1 1 2 2 ...
##  $ class1         : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 2 ...
##  $ level2         : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 1 ...
##  $ level3         : Factor w/ 2 levels "0","1": 1 1 2 1 2 1 1 1 1 1 ...
##  $ bestby         : Factor w/ 2 levels "0","1": 1 1 1 2 2 1 1 2 1 1 ...
##  $ useby          : Factor w/ 2 levels "0","1": 1 1 2 1 1 1 1 1 2 2 ...
##  $ date_days      : int  1 3 1 2 7 1 7 3 1 3 ...
##  $ price          : num  4.79 12.79 4.79 6.39 2.49 ...
##  $ large          : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 2 1 2 1 ...
##  $ age            : num  49.5 49.5 49.5 49.5 49.5 49.5 49.5 49.5 69.5 69.5 ...
##  $ gender         : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
```

```
##  $ educ           : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
##  $ income         : num  6.25 6.25 6.25 6.25 6.25 6.25 6.25 6.25 4.25 4.25 ...
##  $ hhld_size      : int  2 2 2 2 2 2 2 2 1 1 ...
##  $ child_pres     : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
##  $ nonwhite       : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
##  $ preshop_1      : int  4 4 4 4 4 4 4 4 1 1 ...
##  $ preshop_2      : int  4 4 4 4 4 4 4 4 1 1 ...
##  $ preshop_3      : int  4 4 4 4 4 4 4 4 1 1 ...
##  $ preshop_4      : int  4 4 4 4 4 4 4 4 2 2 ...
##  $ instore_5      : int  4 4 4 4 4 4 4 4 1 1 ...
##  $ instore_6      : int  4 4 4 4 4 4 4 4 3 3 ...
##  $ instore_7      : int  4 4 4 4 4 4 4 4 3 3 ...
##  $ tosspre_8      : int  5 5 5 5 5 5 5 5 2 2 ...
##  $ tosspre_9      : int  5 5 5 5 5 5 5 5 1 1 ...
##  $ tosspre_10     : int  5 5 5 5 5 5 5 5 5 5 ...
##  $ tosspre_11     : int  5 5 5 5 5 5 5 5 3 3 ...
##  $ tosspre_12     : int  5 5 5 5 5 5 5 5 5 5 ...
##  $ tosspre_13     : int  5 5 5 5 5 5 5 5 5 5 ...
##  $ tosspre_14     : int  2 2 2 2 2 2 2 2 4 4 ...
##  $ tosspost_15    : int  3 3 3 3 3 3 3 3 4 4 ...
##  $ tosspost_16    : int  3 3 3 3 3 3 3 3 2 2 ...
##  $ tosspost_17    : int  4 4 4 4 4 4 4 4 4 4 ...
##  $ tosspost_18    : int  2 2 2 2 2 2 2 2 5 5 ...
##  $ tosspost_19    : int  2 2 2 2 2 2 2 2 1 1 ...
##  $ amount_20      : int  2 2 2 2 2 2 2 2 2 2 ...
##  $ amount_21      : int  2 2 2 2 2 2 2 2 2 2 ...
##  $ amount_22      : int  2 2 2 2 2 2 2 2 3 3 ...
##  $ amount_23      : int  2 2 2 2 2 2 2 2 0 0 ...
##  $ amount_24      : int  0 0 0 0 0 0 0 0 1 1 ...
##  $ amount_25      : int  2 2 2 2 2 2 2 2 1 1 ...
##  $ amount_26      : int  1 1 1 1 1 1 1 1 2 2 ...
##  $ amount_27      : int  1 1 1 1 1 1 1 1 2 2 ...
##  $ skill_28       : int  5 5 5 5 5 5 5 5 4 4 ...
##  $ skill_29       : int  5 5 5 5 5 5 5 5 3 3 ...
##  $ skill_30       : int  5 5 5 5 5 5 5 5 5 5 ...
##  $ skill_31       : int  5 5 5 5 5 5 5 5 4 4 ...
##  $ skill_32       : int  5 5 5 5 5 5 5 5 5 5 ...
##  $ skill_33       : int  5 5 5 5 5 5 5 5 5 5 ...
##  $ family_34      : int  2 2 2 2 2 2 2 2 4 4 ...
##  $ family_35      : int  5 5 5 5 5 5 5 5 4 4 ...
##  $ compost_36     : int  5 5 5 5 5 5 5 5 5 5 ...
##  $ recycle_37     : int  5 5 5 5 5 5 5 5 1 1 ...
```

```r
data$beef<-factor(data$beef, levels=c(0,1),labels=c("Spinach", "Beef"))
data$class1<-factor(data$class1, levels = c(0,1),labels=c("Planner", "Extemporaneous"))
data$large<-factor(data$large, levels = c(0,1),labels=c("Small", "large"))
data$gender<-factor(data$gender, levels = c(0,1),labels=c("Not Female", "Female"))
data$educ <- factor(data$educ, levels = c(0,1),labels=c("No Degree", "Bachelors or More"))
data$child_pres <- factor(data$child_pres, levels = c(0,1),labels=c("No children", "has children"))
data$nonwhite<-factor(data$nonwhite, levels = c(0,1),labels =c("White", "Not White"))
colSums(is.na(data))
```

```
##       person_id percent_discard            beef          class1          level2
##               0               0               0               0               0
```

```
##          level3          bestby           useby       date_days           price
##               0               0               0               0               0
##           large             age          gender            educ          income
##               0               0               0               0               0
##       hhld_size      child_pres        nonwhite       preshop_1       preshop_2
##               0               0               0               0               0
##       preshop_3       preshop_4       instore_5       instore_6       instore_7
##               0               0              95              16              47
##       tosspre_8       tosspre_9      tosspre_10      tosspre_11      tosspre_12
##               0               0               0               0               0
##      tosspre_13      tosspre_14     tosspost_15     tosspost_16     tosspost_17
##               0               0               0               0               0
##     tosspost_18     tosspost_19       amount_20       amount_21       amount_22
##               0               0               0               0               0
##       amount_23       amount_24       amount_25       amount_26       amount_27
##               0               0               0               0               0
##        skill_28        skill_29        skill_30        skill_31        skill_32
##               0               0               0               0               0
##        skill_33       family_34       family_35      compost_36      recycle_37
##               0               0               0               0               0
```

```r
data2 <- na.omit(data)
colSums(is.na(data2))
```

```
##       person_id percent_discard            beef          class1          level2
##               0               0               0               0               0
##          level3          bestby           useby       date_days           price
##               0               0               0               0               0
##           large             age          gender            educ          income
##               0               0               0               0               0
##       hhld_size      child_pres        nonwhite       preshop_1       preshop_2
##               0               0               0               0               0
##       preshop_3       preshop_4       instore_5       instore_6       instore_7
##               0               0               0               0               0
##       tosspre_8       tosspre_9      tosspre_10      tosspre_11      tosspre_12
##               0               0               0               0               0
##      tosspre_13      tosspre_14     tosspost_15     tosspost_16     tosspost_17
##               0               0               0               0               0
##     tosspost_18     tosspost_19       amount_20       amount_21       amount_22
##               0               0               0               0               0
##       amount_23       amount_24       amount_25       amount_26       amount_27
##               0               0               0               0               0
##        skill_28        skill_29        skill_30        skill_31        skill_32
##               0               0               0               0               0
##        skill_33       family_34       family_35      compost_36      recycle_37
##               0               0               0               0               0
```

```r
set.seed(1234)
n<-nrow(data2)
train_index <- sample(1:n,0.8*n)
train<-data2[train_index,]
test<-data2[-train_index,]
dim(train)
```

```
## [1] 2026   55
```

```r
dim(test)
```

```
## [1] 507  55
```

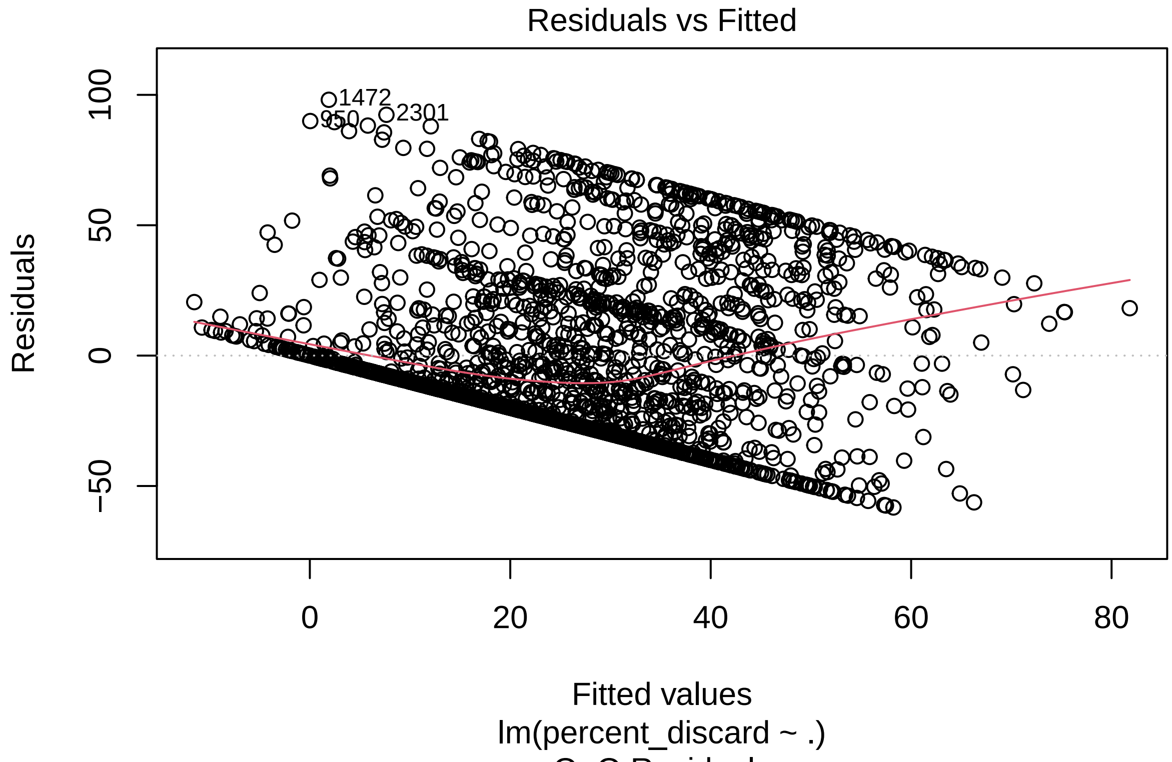
```r
train2<- subset(train,select=-person_id)
test2<- subset(test,select=-person_id)
```

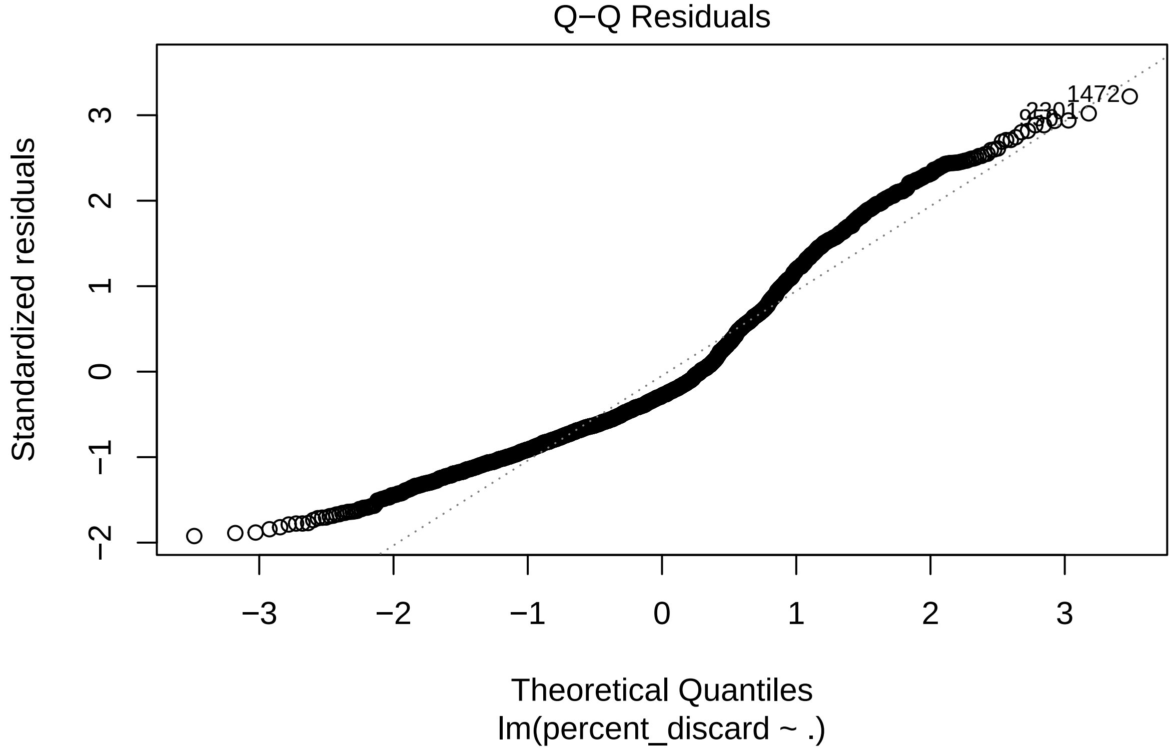
```r
set.seed(1234)
mod<-lm(percent_discard~.,data=train2)
summary(mod)
```

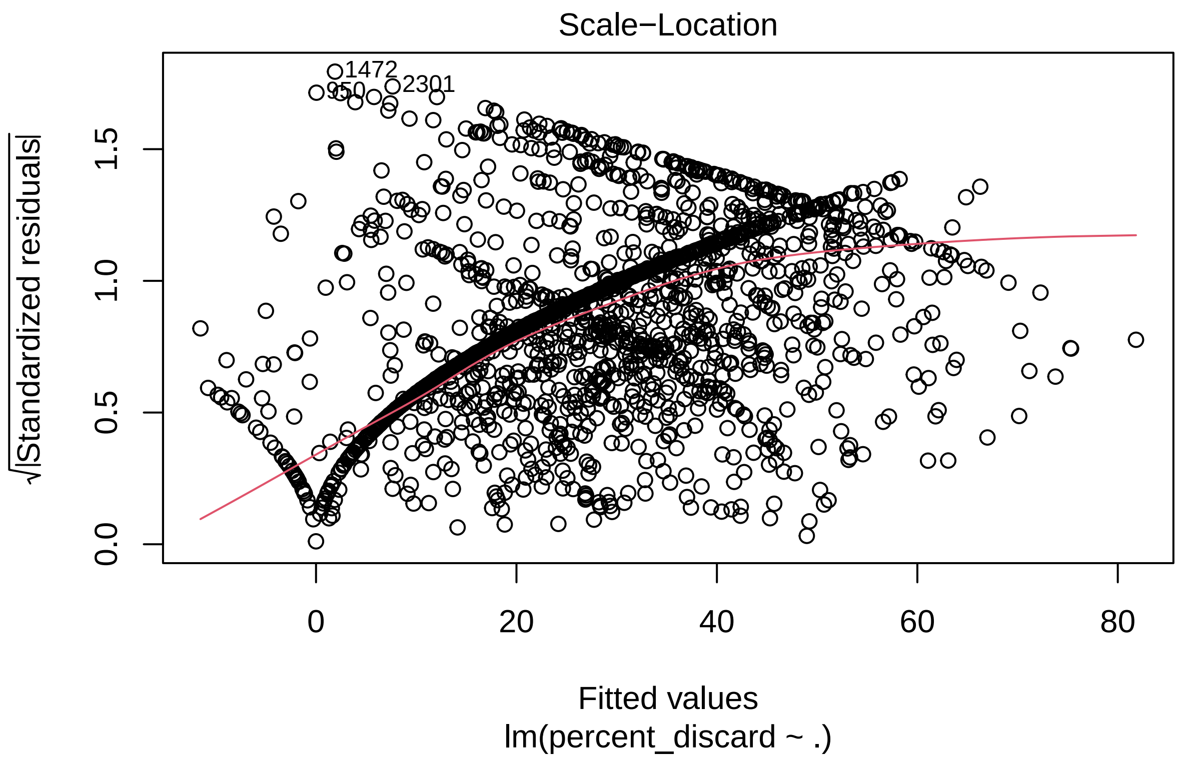
```
##
## Call:
## lm(formula = percent_discard ~ ., data = train2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -58.233 -21.800  -8.585  18.991  98.106
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)               70.60722   12.36528   5.710 1.30e-08 ***
## beefBeef                 -10.69045    2.80361  -3.813 0.000141 ***
## class1Extemporaneous       0.86964    2.40999   0.361 0.718252
## level21                    6.36580    1.72466   3.691 0.000229 ***
## level31                   15.82799    1.73291   9.134  < 2e-16 ***
## bestby1                   -1.72464    1.71724  -1.004 0.315354
## useby1                     2.18661    1.75257   1.248 0.212304
## date_days                 -1.56397    0.36568  -4.277 1.99e-05 ***
## price                      0.16249    0.40828   0.398 0.690682
## largelarge                 3.35382    2.26835   1.479 0.139426
## age                       -0.11488    0.05392  -2.130 0.033258 *
## genderFemale               2.53273    1.63465   1.549 0.121445
## educBachelors or More     -6.01911    1.68648  -3.569 0.000367 ***
## income                    -0.20471    0.14141  -1.448 0.147871
## hhld_size                 -0.76668    0.69614  -1.101 0.270891
## child_preshas children     3.39040    2.14712   1.579 0.114486
## nonwhiteNot White         -5.68814    2.15482  -2.640 0.008362 **
## preshop_1                  0.04020    0.64287   0.063 0.950142
## preshop_2                 -1.46972    0.79180  -1.856 0.063579 .
## preshop_3                 -0.16349    0.85001  -0.192 0.847491
## preshop_4                 -1.28168    0.93130  -1.376 0.168907
## instore_5                  4.00151    0.92234   4.338 1.51e-05 ***
## instore_6                  0.79228    0.80262   0.987 0.323709
## instore_7                 -1.07191    0.75524  -1.419 0.155968
## tosspre_8                 -1.36829    0.74549  -1.835 0.066592 .
## tosspre_9                  0.61909    0.67114   0.922 0.356411
## tosspre_10                -1.79035    0.89579  -1.999 0.045785 *
## tosspre_11                -1.82277    0.69159  -2.636 0.008464 **
## tosspre_12                -1.44962    0.60821  -2.383 0.017247 *
```

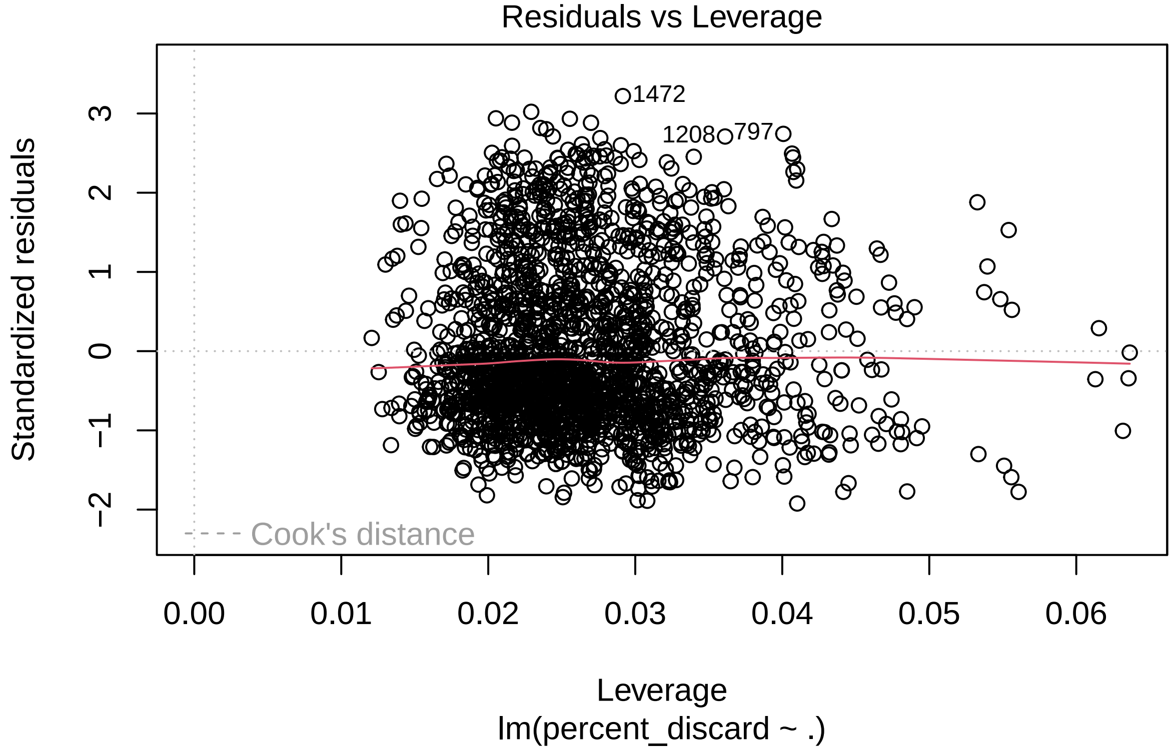
```
## tosspre_13              0.78035    0.67961   1.148 0.251017
## tosspre_14              2.33699    0.80452   2.905 0.003716 **
## tosspost_15             0.14216    0.70461   0.202 0.840126
## tosspost_16             1.32339    0.70928   1.866 0.062215 .
## tosspost_17            -0.57405    0.63868  -0.899 0.368863
## tosspost_18            -0.96515    0.73540  -1.312 0.189534
## tosspost_19            -2.89009    0.72781  -3.971 7.42e-05 ***
## amount_20              -1.20062    1.01025  -1.188 0.234803
## amount_21               2.55440    1.02351   2.496 0.012652 *
## amount_22               2.87670    0.96727   2.974 0.002975 **
## amount_23               0.91243    0.76846   1.187 0.235235
## amount_24              -0.03818    0.73664  -0.052 0.958674
## amount_25               3.53904    1.26611   2.795 0.005237 **
## amount_26              -0.11883    1.15301  -0.103 0.917925
## amount_27              -5.56780    1.28725  -4.325 1.60e-05 ***
## skill_28               -2.59702    1.02266  -2.539 0.011178 *
## skill_29               -0.71768    0.93871  -0.765 0.444638
## skill_30               -0.94944    1.06159  -0.894 0.371237
## skill_31                2.24031    1.22101   1.835 0.066687 .
## skill_32                1.26550    1.35943   0.931 0.352014
## skill_33               -2.06846    1.57022  -1.317 0.187890
## family_34               0.29063    0.81444   0.357 0.721246
## family_35              -2.32365    0.82304  -2.823 0.004802 **
## compost_36              1.20678    0.90585   1.332 0.182946
## recycle_37             -1.67991    0.54273  -3.095 0.001994 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 30.93 on 1972 degrees of freedom
## Multiple R-squared:  0.1861, Adjusted R-squared:  0.1643
## F-statistic: 8.509 on 53 and 1972 DF,  p-value: < 2.2e-16
```

```r
par(mfrow=c(1,1))
plot(mod)
```


Residuals vs Fitted
Residuals
100
50
0
−50
0
20
40
60
80
1472
950
2301
Fitted values
lm(percent_discard ~ .)


Q−Q Residuals
Standardized residuals
3
2
1
0
−1
−2
−3
−2
−1
0
1
2
3
1472
2301
Theoretical Quantiles
lm(percent_discard ~ .)






```r
confint(mod)
```

```
##                              2.5 %       97.5 %
## (Intercept)             46.35683780 94.857605776
## beefBeef               -16.18880930 -5.192098079
## class1Extemporaneous    -3.85675252  5.596030680
```

```
## level21                    2.98346367  9.748145743
## level31                   12.42946664 19.226511092
## bestby1                   -5.09243635  1.643159310
## useby1                    -1.25046951  5.623686285
## date_days                 -2.28114102 -0.846802691
## price                     -0.63821405  0.963194935
## largelarge                -1.09479436  7.802431565
## age                       -0.22062865 -0.009128154
## genderFemale              -0.67309015  5.738559485
## educBachelors or More     -9.32658134 -2.711639186
## income                    -0.48203581  0.072613900
## hhld_size                 -2.13192580  0.598574178
## child_preshas children    -0.82046679  7.601275755
## nonwhiteNot White         -9.91409901 -1.462190634
## preshop_1                 -1.22056928  1.300974400
## preshop_2                 -3.02257511  0.083140919
## preshop_3                 -1.83049867  1.503508996
## preshop_4                 -3.10812269  0.544753959
## instore_5                  2.19265514  5.810368008
## instore_6                 -0.78179977  2.366360070
## instore_7                 -2.55305762  0.409233467
## tosspre_8                 -2.83030814  0.093734420
## tosspre_9                 -0.69713057  1.935303662
## tosspre_10                -3.54713294 -0.033559010
## tosspre_11                -3.17909379 -0.466442362
## tosspre_12                -2.64241825 -0.256818652
## tosspre_13                -0.55249058  2.113180984
## tosspre_14                 0.75918834  3.914794856
## tosspost_15               -1.23969788  1.524021615
## tosspost_16               -0.06763022  2.714411781
## tosspost_17               -1.82661315  0.678505818
## tosspost_18               -2.40739020  0.477095860
## tosspost_19               -4.31744442 -1.462726833
## amount_20                 -3.18189587  0.780650207
## amount_21                  0.54711234  4.561677847
## amount_22                  0.97971038  4.773680729
## amount_23                 -0.59465388  2.419512066
## amount_24                 -1.48284675  1.406495282
## amount_25                  1.05599253  6.022091727
## amount_26                 -2.38007106  2.142409903
## amount_27                 -8.09231481 -3.043287566
## skill_28                  -4.60263652 -0.591406380
## skill_29                  -2.55863946  1.123284410
## skill_30                  -3.03139472  1.132505929
## skill_31                  -0.15430892  4.634920534
## skill_32                  -1.40056003  3.931569330
## skill_33                  -5.14792356  1.011011759
## family_34                 -1.30663160  1.887893470
## family_35                 -3.93776333 -0.709528325
## compost_36                -0.56973988  2.983291322
## recycle_37                -2.74429445 -0.615519922
```

```r
library(car)
```

```
## Loading required package: carData
```

```r
vif(mod)
```

```
##        beef      class1      level2      level3      bestby       useby
##    4.087858    3.016423    1.172878    1.158338    1.393017    1.435828
##   date_days       price       large         age      gender        educ
##    1.253473    5.701405    2.724981    1.579481    1.224328    1.327541
##      income   hhld_size  child_pres    nonwhite   preshop_1   preshop_2
##    1.714031    2.025995    1.832400    1.220268    1.458036    1.985961
##   preshop_3   preshop_4   instore_5   instore_6   instore_7   tosspre_8
##    2.120233    1.736466    2.170858    1.666638    1.588028    1.839133
##   tosspre_9  tosspre_10  tosspre_11  tosspre_12  tosspre_13  tosspre_14
##    1.635441    1.625348    1.579954    1.523245    1.623756    1.626493
## tosspost_15 tosspost_16 tosspost_17 tosspost_18 tosspost_19   amount_20
##    1.921288    1.651836    1.505534    1.712378    1.520100    2.085997
##   amount_21   amount_22   amount_23   amount_24   amount_25   amount_26
##    2.096782    1.832154    1.528123    1.580475    2.329269    2.059296
##   amount_27    skill_28    skill_29    skill_30    skill_31    skill_32
##    1.903086    1.818447    1.727206    1.652142    1.974089    2.253229
##    skill_33   family_34   family_35  compost_36  recycle_37
##    2.104551    1.693586    1.670961    1.361420    1.257829
```

```r
set.seed(1234)
#lm test pred
pred<- predict(mod, newdata=test2)
rmse<- sqrt(mean((test2$percent_discard-pred)^2))
mae<-mean(abs(test2$percent_discard -pred))
#r^2 here!
ss_tot<- sum((test2$percent_discard-mean(test2$percent_discard))^2)
ss_resi<- sum((test2$percent_discard -pred)^2)
r2<-1-(ss_resi /ss_tot)

rmse
```

```
## [1] 28.68509
```

```r
mae
```

```
## [1] 23.47396
```

```r
r2
```

```
## [1] 0.2095788
```

```r
#rand forest model
library(randomForest)
```

```
## randomForest 4.7-1.2
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

```r
set.seed(1234)
rf<-randomForest(percent_discard~.,data=train2,ntree=500, importance=TRUE)
print(rf)
```

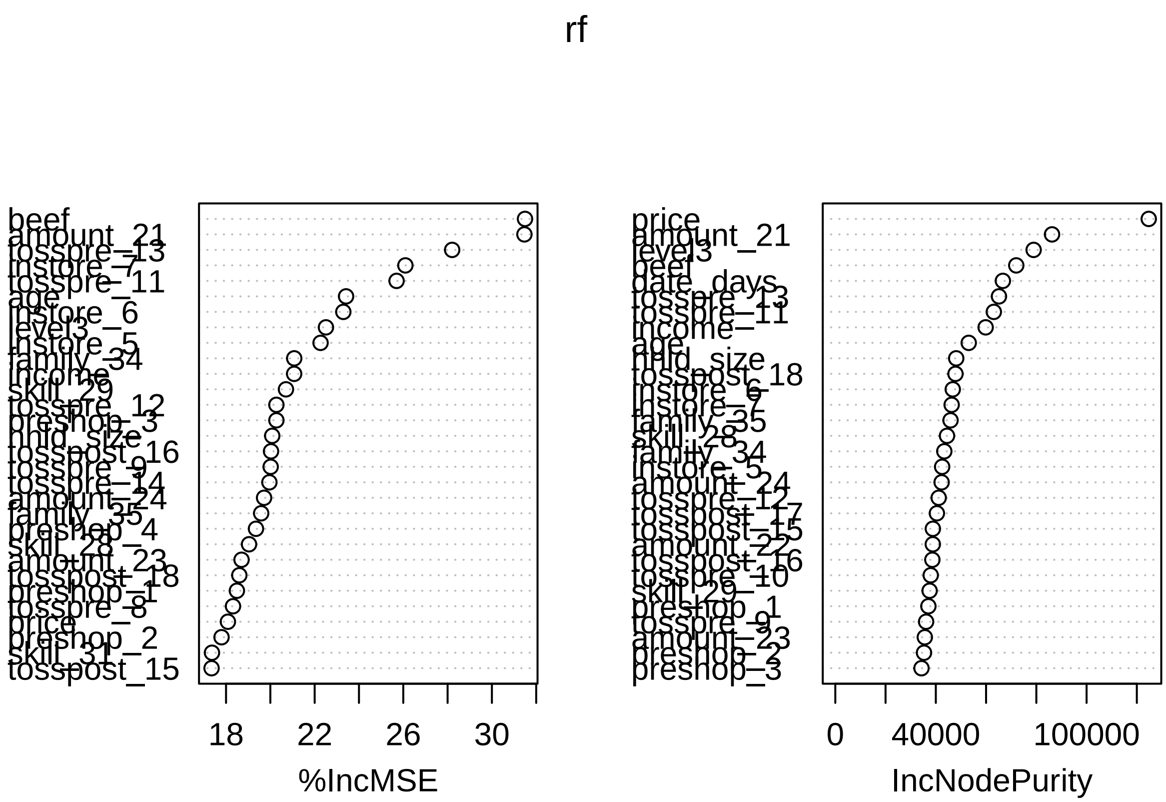
```
##
## Call:
##  randomForest(formula = percent_discard ~ ., data = train2, ntree = 500,      importance = TRUE)
##                Type of random forest: regression
##                      Number of trees: 500
## No. of variables tried at each split: 17
##
##           Mean of squared residuals: 595.9989
##                     % Var explained: 47.89
```

```r
importance(rf)
```

```
##                %IncMSE IncNodePurity
## beef        31.4940427      72013.99
## class1       8.8759087      12059.44
## level2      -3.1424837      27665.82
## level3      22.5094419      78931.93
## bestby      -4.0488355      24432.92
## useby       -0.5914817      26842.29
## date_days    2.7589030      66685.53
## price       18.0835203     124733.57
## large        2.9303715      32776.45
## age         23.4180153      53101.58
## gender      13.4903299      17401.19
## educ        15.6563726      29907.58
## income      21.0737228      59834.47
## hhld_size   20.0833091      48121.38
## child_pres  12.8290709      12904.74
## nonwhite    10.6010305      10132.95
## preshop_1   18.4898752      37049.54
## preshop_2   17.7968463      35308.94
## preshop_3   20.2727685      34302.15
## preshop_4   19.3508838      31476.84
## instore_5   22.2667693      42522.68
## instore_6   23.2931767      46740.96
## instore_7   26.0949313      46301.25
## tosspre_8   18.3134997      33425.77
## tosspre_9   20.0140855      36144.67
## tosspre_10  17.1956704      37979.26
## tosspre_11  25.6974199      63084.26
## tosspre_12  20.2728932      41151.03
## tosspre_13  28.2058742      65138.30
## tosspre_14  19.9566151      30556.18
## tosspost_15 17.3461324      38821.41
## tosspost_16 20.0303335      38629.18
## tosspost_17 15.4627428      40432.67
## tosspost_18 18.5990493      47827.14
## tosspost_19 16.8595453      33535.54
```

```
## amount_20    15.5231809      33956.84
## amount_21    31.4676074      86252.68
## amount_22    16.2754262      38809.41
## amount_23    18.6997356      35626.04
## amount_24    19.7163766      42321.17
## amount_25    15.9060637      24595.92
## amount_26    14.0070552      21033.55
## amount_27    12.9102961      18624.33
## skill_28     19.0382100      44454.20
## skill_29     20.7046972      37529.78
## skill_30     16.1520818      28663.60
## skill_31     17.3640796      28170.49
## skill_32     14.9969236      21597.80
## skill_33     14.0945035      22134.40
## family_34    21.0771436      43377.27
## family_35    19.5856096      45864.06
## compost_36   15.8684931      25625.79
## recycle_37   15.7659667      33137.08
```

```r
varImpPlot(rf)
```




```r
pred_rf<-predict(rf,newdata=test2)
rmse_rf<- sqrt(mean((test2$percent_discard-pred_rf)^2))
mae_rf<-mean(abs(test2$percent_discard -pred_rf))
#r^2 here!
ss_tot2<- sum((test2$percent_discard-mean(test2$percent_discard))^2)
ss_resi2<- sum((test2$percent_discard -pred_rf)^2)
r2_rf<-1-(ss_resi2 /ss_tot2)
```

```
rmse_rf
```

```
## [1] 23.7777
```

```
mae_rf
```

```
## [1] 18.38851
```

```
r2_rf
```

```
## [1] 0.4568925
```

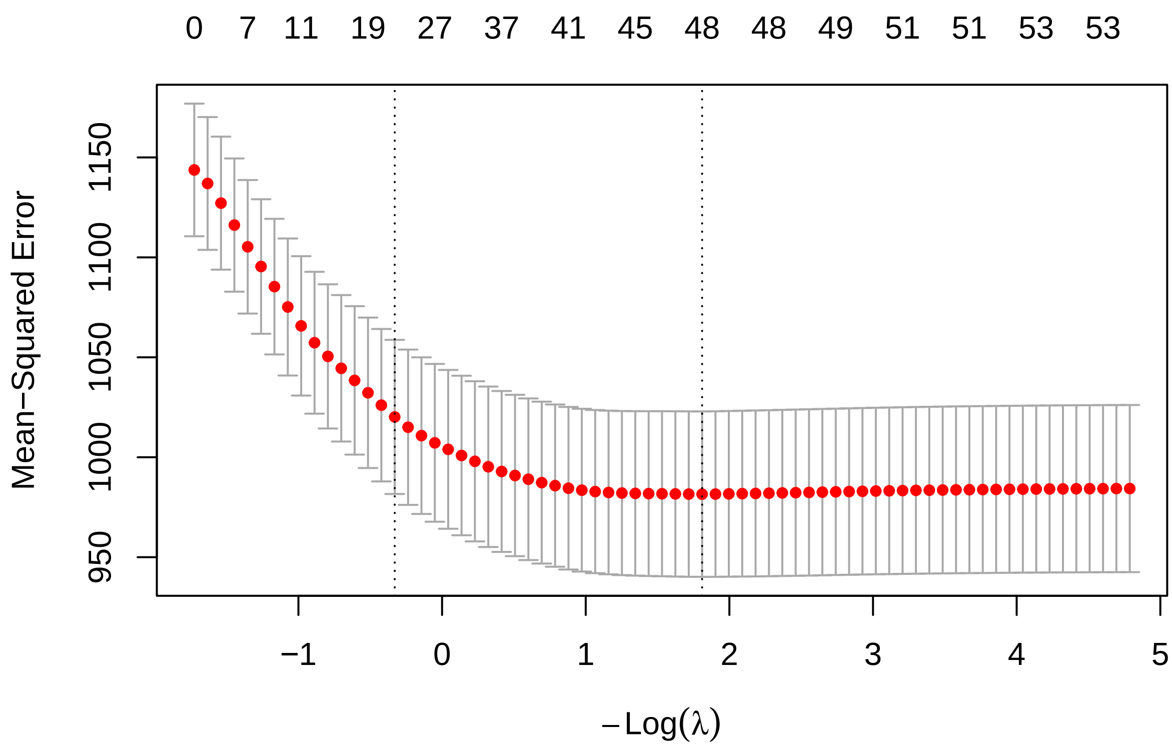
```
#Lasso
set.seed(1234)
library(glmnet)
```

```
## Loading required package: Matrix
```

```
## Loaded glmnet 4.1-10
```

```
x_train<-model.matrix(percent_discard~.,train2)[,-1]
y_train <-train2$percent_discard
x_test<-model.matrix(percent_discard~.,test2)[,-1]
y_test<-test2$percent_discard

las<-cv.glmnet( x_train,y_train, alpha=1)
plot(las)
```



```r
las$lambda.min
```

```
## [1] 0.1636193
```

```r
(lam<-coef(las,s="lambda.min")) #coef at opt lam
```

```
## 54 x 1 sparse Matrix of class "dgCMatrix"
##                          lambda.min
## (Intercept)             69.86567176
## beefBeef                -9.21607534
## class1Extemporaneous     0.90124273
## level21                  5.74892180
## level31                 15.30589920
## bestby1                 -1.52947939
## useby1                   1.86184870
## date_days               -1.46205911
## price                    .
## largelarge               3.68472655
## age                     -0.09120415
## genderFemale             1.99635227
## educBachelors or More   -6.13106428
## income                  -0.17560473
## hhld_size               -0.21915467
## child_preshas children   2.03360053
## nonwhiteNot White       -4.67419047
## preshop_1                .
## preshop_2               -1.20024232
## preshop_3               -0.13366661
## preshop_4               -0.98767942
## instore_5                3.43708522
## instore_6                0.54271182
## instore_7               -0.78095013
## tosspre_8               -0.88428761
## tosspre_9                0.34666425
## tosspre_10              -1.71375475
## tosspre_11              -1.60242567
## tosspre_12              -1.15696659
## tosspre_13               0.36187598
## tosspre_14               2.02469778
## tosspost_15              .
## tosspost_16              0.98330248
## tosspost_17             -0.60976990
## tosspost_18             -0.79761186
## tosspost_19             -2.61422786
## amount_20               -0.92387546
## amount_21                2.51034017
## amount_22                2.62002987
## amount_23                0.42931662
## amount_24                .
## amount_25                2.99990021
## amount_26                .
## amount_27               -4.86893462
## skill_28                -2.68528036
```

```
## skill_29               -0.34929659
## skill_30               -0.79643200
## skill_31                1.76901234
## skill_32                0.38006256
## skill_33               -1.17461392
## family_34               0.03263354
## family_35              -2.06041968
## compost_36              0.73233289
## recycle_37             -1.58613262
```

```
set.seed(1234)
library(glmnet)
#lambda matrix here
#(lam_mat<-as.matrix(lam))
lam_mat<-as.matrix(lam)
lam_df <-data.frame(variable =rownames(lam_mat),coefficient=lam_mat[, 1])
lam_df<-lam_df[order(-abs(lam_df$coefficient)),]
#top 10 coef
head(lam_df[lam_df$coefficient!= 0,], 10)
```

```
##                                    variable coefficient
## (Intercept)                     (Intercept)   69.865672
## level31                             level31   15.305899
## beefBeef                           beefBeef   -9.216075
## educBachelors or More educBachelors or More   -6.131064
## level21                             level21    5.748922
## amount_27                         amount_27   -4.868935
## nonwhiteNot White         nonwhiteNot White   -4.674190
## largelarge                       largelarge    3.684727
## instore_5                         instore_5    3.437085
## amount_25                         amount_25    2.999900
```

```
(sum(lam_df$coefficient[-1]!=0))
```

```
## [1] 48
```

```
(nrow(lam_df)-1)
```

```
## [1] 53
```

```
pred_las<-predict(las,s="lambda.min", newx=x_test)
rmse_las<- sqrt(mean((y_test-pred_las)^2))
mae_las<-mean(abs(y_test -pred_las))
#r^2 here!
ss_tot3<- sum((y_test-mean(test2$percent_discard))^2)
ss_resi3<- sum((y_test-pred_las)^2)
r2_las<-1-(ss_resi3 /ss_tot3)

rmse_las
```

```
## [1] 28.6694
```

```r
mae_las
```

```
## [1] 23.56485
```

```r
r2_las
```

```
## [1] 0.210443
```